ECE 501 - DIGITAL IMAGE PROCESSING

Explainable Multi-Label Content-Based Image Retrieval for Aerial Images using Classical Computer Vision

Weekly Progress Report

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1 Introduction

During Week 6, the team continued working on Phase 5 of the proposed Content-Based Image Retrieval (CBIR) system. The previous phase focused on identifying possible improvements to the existing retrieval pipeline, particularly in feature fusion, explainability, uncertainty analysis, and performance evaluation.

This week, the team continued discussing these improvements and considered how they could be incorporated into the existing system. Along with the ongoing work on Phase 5, initial discussions were started regarding the next phase of the project. The aim was to organize the remaining work and identify the steps required to move towards a more interpretable and reliable aerial-image retrieval system.

2 Continuation of Phase 5

Phase 5 builds upon the existing retrieval framework, which uses handcrafted colour, texture, shape, and spatial features to compare aerial images. The focus remains on improving the usefulness and interpretability of the retrieval results without introducing machine learning or deep learning methods.

The team continued reviewing the proposed improvements and considering how each component could be developed and evaluated within the current pipeline.

2.1 Feature-Weight Tuning

Different feature descriptors capture different visual characteristics of an aerial image. Their contributions to the final similarity score may vary depending on the query and the type of scene being retrieved.

The team continued considering how feature-fusion weights could be selected using a validation split. The plan is to compare different weight combinations and observe their effect on retrieval metrics. This would help determine whether assigning different importance to individual feature cues can improve the ranking of relevant images.

2.2 Explainability of Retrieval Results

The team continued discussing ways to make the retrieval process easier to understand. The proposed approach involves displaying the contribution of individual feature cues to the final similarity score.

For example, a retrieved image may have a high colour similarity but a lower shape similarity. Presenting these individual scores could help explain why the image received its final ranking.

Region-based visualizations and heat maps are also being considered as possible ways to highlight image areas that may contribute to visual similarity. These explanations are intended to make the retrieval process more transparent to users.

2.3 Uncertainty Analysis

Another area of continued discussion was the identification of potentially unreliable retrieval results. The proposed approach involves examining the difference between the similarity scores of the top-ranked images and checking for disagreement among individual feature cues.

For instance, if two candidate images have very similar overall scores, the system may have difficulty distinguishing between them. Similarly, if colour and texture cues indicate different levels of similarity, the result may require further examination.

The team plans to investigate these indicators to determine whether they can help identify uncertain retrievals.

3 Initial Planning for the Next Phase

Alongside the continuation of Phase 5, the team began discussing the next phase of the project. The focus of these discussions was to identify how the planned improvements could be organized into an implementation and evaluation workflow.

The next phase is expected to build upon the work completed so far and move towards integrating the selected improvements into the existing retrieval system. Particular attention will be given to maintaining consistency with the current feature-extraction and similarity-computation pipeline.

The following activities were identified for further planning:

- Identify the Phase 5 components that should be implemented and tested first.
- Plan how feature-weight tuning can be integrated into the existing score-fusion approach.
- Determine how individual feature contributions can be displayed alongside retrieval results.
- Explore practical ways to represent uncertainty using score margins and feature-cue disagreement.
- Organize the evaluation process using held-out query images.
- Define a sequence of implementation, testing, and comparison activities for the upcoming phase.

4 Evaluation Considerations

The team also considered how the planned improvements should be evaluated. Precision@K and mean Average Precision (mAP) will remain important metrics for assessing the quality of the retrieval rankings. Multi-label relevance will also be examined using Jaccard similarity thresholds.

The evaluation will help determine whether changes to feature fusion improve retrieval performance. It will also be important to assess whether the proposed explanations provide useful information about the retrieved images.

The planned evaluation will use held-out queries to assess the system independently of the queries used for tuning. The results can then be compared with the existing retrieval approach to identify improvements and limitations.

5 Work Completed

During Week 6, the following activities were carried out:

- Continued work and discussions related to Phase 5.
- Reviewed the proposed feature-weight tuning approach.
- Continued planning the feature-wise score breakdown and region-based visualizations.
- Further discussed uncertainty indicators based on similarity-score margins and feature-cue disagreement.
- Started planning the implementation and evaluation workflow for the next phase.
- Identified key activities required to integrate and assess the proposed improvements.

6 Plan for Next Week

The following activities are planned for the upcoming week:

- Continue the development of Phase 5 components.
- Finalize the implementation sequence for the planned improvements.
- Begin implementing the selected components, as appropriate.
- Explore feature-weight combinations using validation queries.
- Plan the integration of retrieval explanations and uncertainty indicators.
- Prepare the evaluation procedure for comparing the updated system with the existing approach.

7 Conclusion

During Week 6, the team continued working on Phase 5 and further considered improvements to the existing CBIR system. Discussions focused on feature-weight tuning, explainability, uncertainty analysis, and evaluation. The team also began planning the next phase, with the objective of organizing the remaining implementation and testing activities.

The upcoming work will focus on progressing from planning towards implementation and systematically evaluating the proposed improvements while retaining the project's emphasis on classical computer vision and handcrafted image features.